

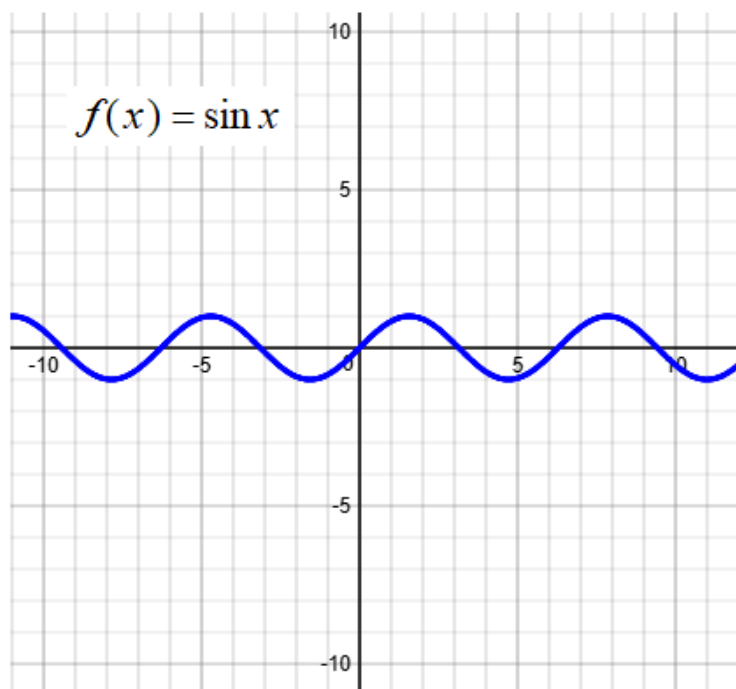
Calculus I Lesson 29C

1) Find $\lim_{x \rightarrow \infty} \sin x$

a) $\underline{\quad ? \quad} \leq \sin x \leq \underline{\quad ? \quad}$

b) $\sin(10) = \underline{\quad ? \quad}$ $\sin(100) = \underline{\quad ? \quad}$ $\sin(1000) = \underline{\quad ? \quad}$

c) $\lim_{x \rightarrow \infty} \sin x = \underline{\quad ? \quad}$

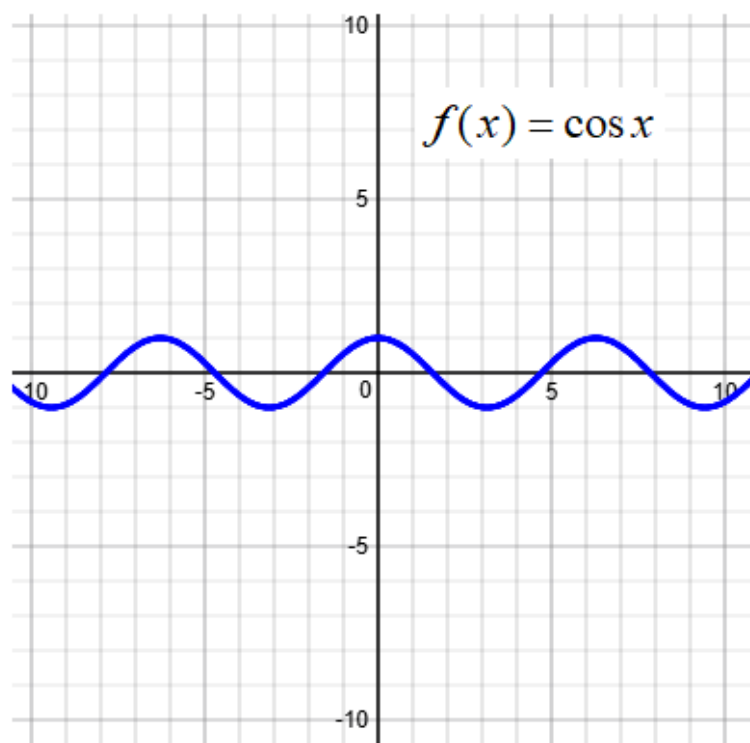


2) Find $\lim_{x \rightarrow \infty} \cos x$

a) $\underline{\quad ? \quad} \leq \cos x \leq \underline{\quad ? \quad}$

b) $\cos(10) = \underline{\quad ? \quad}$ $\cos(100) = \underline{\quad ? \quad}$ $\cos(1000) = \underline{\quad ? \quad}$

c) $\lim_{x \rightarrow \infty} \cos x = \underline{\quad ? \quad}$



3) Find $\lim_{x \rightarrow \infty} \sin \frac{1}{x}$

a) $\lim_{x \rightarrow \infty} \frac{1}{x} = \underline{\hspace{1cm}} ?$

b) $\lim_{x \rightarrow \infty} \sin \frac{1}{x} = \underline{\hspace{1cm}} ?$

4) Find $\lim_{x \rightarrow \infty} \cos \frac{1}{x}$

a) $\lim_{x \rightarrow \infty} \frac{1}{x} = \underline{\hspace{1cm} ? \hspace{1cm}}$

b) $\lim_{x \rightarrow \infty} \cos \frac{1}{x} = \underline{\hspace{1cm} ? \hspace{1cm}}$

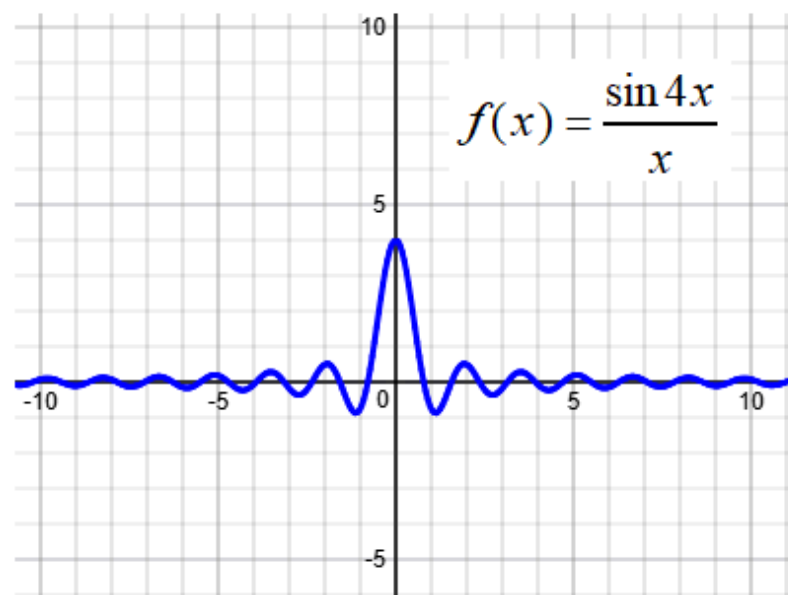
5) Find $\lim_{x \rightarrow \infty} \frac{\sin 4x}{x}$

Hint: $-1 \leq \sin 4x \leq 1$ and $\frac{-1}{x} \leq \frac{\sin 4x}{x} \leq \frac{1}{x}$

Note: x is positive since $x \rightarrow \infty$

a) $\lim_{x \rightarrow \infty} \frac{-1}{x} = \underline{\quad ? \quad}$ and $\lim_{x \rightarrow \infty} \frac{1}{x} = \underline{\quad ? \quad}$

b) $\lim_{x \rightarrow \infty} \frac{\sin 4x}{x} = \underline{\quad ? \quad}$

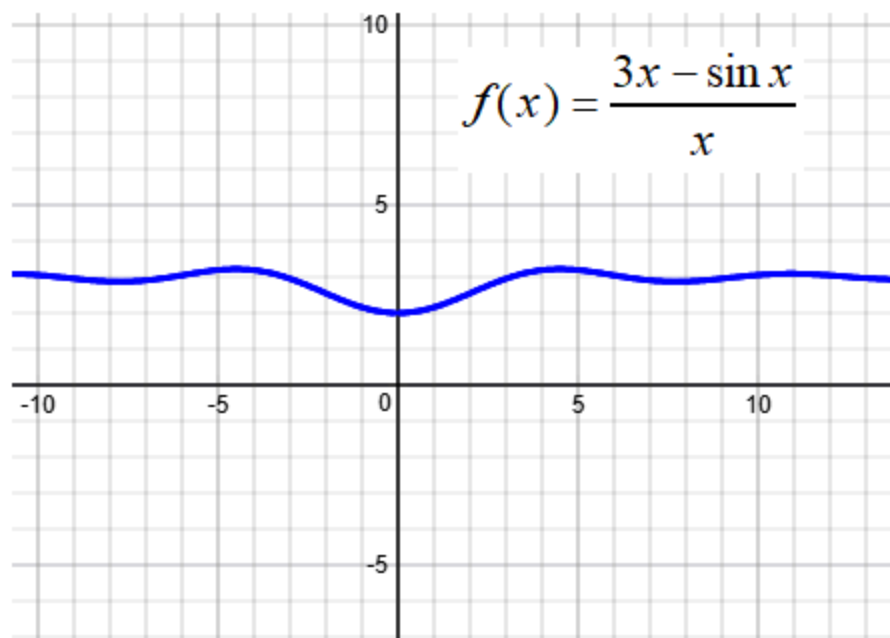


6) Find $\lim_{x \rightarrow \infty} \frac{3x - \sin x}{x}$

Hint: $-1 \leq \sin x \leq 1$ and $\frac{-1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$

a) Simplify $\frac{3x - \sin x}{x} = \frac{3x}{x} - \frac{\sin x}{x} = \underline{\hspace{1cm}} ?$

b) $\lim_{x \rightarrow \infty} \frac{3x - \sin x}{x} = \underline{\hspace{1cm}} ?$



6) Find $\lim_{x \rightarrow \infty} \frac{3x - \sin x}{x}$

Hint: $-1 \leq \sin x \leq 1$ and $\frac{-1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$

a) Simplify $\frac{3x - \sin x}{x} = \frac{3x}{x} - \frac{\sin x}{x} = \underline{\hspace{1cm}} ?$

b) $\lim_{x \rightarrow \infty} \frac{3x - \sin x}{x} = \underline{\hspace{1cm}} ?$

